Federated Learning for EEG-Based Working-Memory Decoding

Model Architecture and Empirical Evaluation (Deliverable 2)

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Abstract

We investigate EEG-based working-memory load decoding with centralized, federated and personalized learning on OpenNeuro `ds003655`. A two-participant preprocessing pilot was expanded to 24 participants. The primary dataset comprised 2,878 trials and 57 delay-period band-power features per trial. Three neural-network training strategies were evaluated on 359 matched trials from four held-out participants. Centralized, FedAvg and personalized FedAvg macro-F1 scores were 0.3036, 0.3276 and 0.3397, respectively. All three ROC-AUC scores were near 0.50, indicating limited discrimination on this split.

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1 Phase 1: Problem Definition

1.1 Problem Statement

Working memory is the short-term maintenance and manipulation of information, and it is central to reasoning and learning. EEG measures brain activity non-invasively and cheaply, so decoding working-memory demand from EEG is of interest for cognitive-state monitoring and brain–computer interfaces. Two practical obstacles limit this:

1. **Inter-subject variability.** EEG differs strongly between people (anatomy, electrode contact, noise, individual rhythm frequencies), so a model trained on some people often degrades on others, and data from many participants is needed.
2. **Privacy and data silos.** EEG is sensitive personal data. Pooling recordings from different labs or clinics is often restricted by consent and regulation.

Federated learning trains a shared model by exchanging model updates instead of raw recordings, which addresses the second obstacle. However, each participant's data is distributed differently (non-IID), which is exactly the first obstacle and is known to make federated optimisation harder [3]. This project studies that interaction for EEG working-memory decoding.

1.2 Research Questions

1. How does a federated model compare with a centralized baseline for working-memory condition decoding?
2. How sensitive are results to participant-level distribution differences?
3. Does a clearly defined personalization strategy improve cross-subject generalization?

These are planned experiments, not results at this stage.

1.3 Target Audience and Research Gap

This is an original research (R&D) project. The intended audience is researchers in neuroengineering, brain–computer interfaces and privacy-preserving machine learning. The gap is a reproducible, small-scale comparison of centralized, federated and personalized learning for EEG working-memory decoding under non-IID, cross-subject conditions, in which data quality and label validity are examined rather than assumed.

Scope. The initial pilot uses two participants. The later modeling experiment expands to 24 participants.

1.4 Justification for Machine Learning

A rule-based program would need hand-written rules from EEG measurements to working-memory demand. This is unsuitable because:

- **High-dimensional, weak, distributed signal.** Each trial is a 19-channel time series at 500 Hz; memory-load effects appear as subtle spectral changes spread over channels, with no single threshold separating conditions.
- **Subject-specific patterns.** The same state looks different across people, so fixed rules do not transfer, whereas learned models can adapt to these differences.
- **Low signal-to-noise ratio.** Artifacts (eye movement, muscle activity, amplifier saturation; see Section 3) require statistical models that weigh evidence over many features.
- **The research question itself is about learning.** Federated optimisation and personalization are properties of learning algorithms and have no rule-based counterpart.

2 Phase 2: Data Acquisition

2.1 Dataset and Collection

The data are from OpenNeuro dataset `ds003655`, task `VerbalWorkingMemory` (`https://openneuro.org/datasets/ds003655`), a modified Sternberg task with retention (R) and manipulation (M) conditions at set sizes 5, 6 and 7, giving the conditions 5R, 6R, 7R, 5M, 6M and 7M. Two participants, `sub-001` and `sub-002`, were downloaded from the public repository. The local inventory (`reports/download_inventory.csv`, produced by `01_check_downloads.py`) records eight files per participant and about 269.05 MB in total. Raw files are excluded from Git through `.gitignore`. No download API log was captured.

Table 1: Recording metadata.

Property	sub-001	sub-002
Sampling rate	500 Hz	500 Hz
Channels recorded	21	21
EEG channels	19	19
EOG channels	EOGv, EOGh	EOGv, EOGh
Duration	3077 s	3035 s

The EEG channels are Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, T5, P3, Pz, P4, T6, O1, O2.

2.2 Trial Structure

`04_build_trial_table.py` builds `reports/trial_table.csv` from the event markers: 240 trials (120 per participant), 20 per condition per participant. Each row holds the onsets of the baseline, encoding, delay and probe phases, the response information and quality flags. The delay-to-probe interval is about 6.7 s, but all scripts use the actual event onsets.

3 Phase 2: Validation and Data-Centric Cleaning

3.1 File and Event Validation

Both recordings loaded correctly, with 120 trial starts and 20 baseline, encoding, delay and probe markers per condition. The event tables had no exact duplicate rows and no onsets outside the recording. These checks confirm structure, not scientific validity.

3.2 Response-Label Uncertainty

There are 244 response markers for 240 probes; some are late, duplicated or ambiguous. The trial table flags eight trials for response ambiguity (6 in sub-001, 2 in sub-002), and the later probe-based audit marks 225 of 240 rows as PASS and 15 as REVIEW; the two counts come from different checks and are to be reconciled. Late responses are judged with a five-second heuristic that is not a validated rule. Uncertain cases are kept, not silently resolved, and the event-to-label mapping is provisional; no supervised model will be trained before the labels are validated.

3.3 Channel-Type Correction

MNE initially labelled `EOGv` and `EOGh` as EEG. They are now set to type `eog` before channel selection, so features use only the 19 EEG channels (`12_inspect_channel_types.py`, the Colab

pilot notebook).

3.4 Initial Signal-Quality Findings

sub-001 boundary values. In all 21 channels the minimum value reaches about $-10{,}239.69\,\mu$V (maximum absolute value $10{,}239.69\,\mu$V). Between 332.5 and 333.5 s each channel has many samples near this lower boundary, including consecutive runs of about 0.48–0.52 s, and the same boundary was not observed in sub-002 in that interval. This is suspicious for clipping, saturation or a data-representation issue, but **the cause is not confirmed** (`reports/signal_quality_decisions.md`).

Amplitude screening before correction. Each trial window was flagged if its maximum channel peak-to-peak or absolute amplitude exceeds a heuristic threshold of $500\,\mu$V.

Table 2: Heuristic amplitude flags per trial window (not validated rejection rules).

Participant	No flag	Peak-to-peak only	Both flags
sub-001	112	6	2
sub-002	0	0	120
Total	112	6	122

Table 3: Window amplitude by flag group (μV).

Group	Median p–p	Mean p–p	Median abs. amp.
No flag	406.66	411.35	258.31
Peak-to-peak only	524.28	550.06	371.22
Both flags	2550.53	2588.39	1325.66

Every sub-002 trial is flagged, which does not by itself mean every trial should be discarded. Whole-record channel ranges in sub-002 are large but vary by channel (e.g. Pz 5461.00, T3 5318.31, O1 5506.38 μV), and the EOG channels also have large extrema. To tell a single bad channel from a participant-wide difference, the per-channel, per-trial audit (`24_inspect_trial_channel_amplitudes.py`, output `reports/trial_channel_amplitude_audit.csv`) was computed for every delay-to-probe window (Table 4).

Table 4: Per-trial channel amplitude audit by participant (120 windows each; μV).

Quantity	sub-001	sub-002
Max channel peak-to-peak, median (range)	410.8 (317.4–1612.2)	2551.4 (2444.9–2989.5)
Max absolute amplitude, median	260.5	1325.7
Median-over-channels peak-to-peak, mean	256.5	613.0
Channels above 500 μV p–p per window, mean (range)	0.24 (0–19)	13.2 (12–19)
Channel with the largest p–p	Fp2 50, T3 45, O2 24, F8 1 trials	T3 in all 120 trials

Two observations follow. First, the sub-002 difference is not confined to one channel: the

typical channel is itself about 2.4 times larger than in sub-001, and 12 or more of the 19 EEG channels exceed 500 μV in every window. Second, T3 has the largest range in every sub-002 window, so a channel-specific problem may be superimposed on the general difference. In sub-001 the flags are rare and concentrated in a few windows (the maximum of 19 channels over threshold belongs to a single window). The cause of the sub-002 pattern (recording or reference differences, scaling, drift, or genuine activity) is **still not established**. Note also that even the unflagged sub-001 windows have a median channel peak-to-peak range of roughly 250 μV, which is large for EEG; we hypothesise that this reflects the absence of high-pass filtering, but this has not been tested.

3.5 Signal Correction and Final Pilot Audit

The pilot notebook cropped each recording from 10 s before the first delay to 10 s after the last probe, within recording bounds. The suspicious sub-001 interval preceded this crop. Processing applied a 50 Hz IIR notch, a 1–40 Hz fourth-order Butterworth IIR band-pass, and resampling from 500 to 100 Hz. The linked-earlobe reference was retained.

FastICA was fitted separately to the EEG channels of each participant using a 0.99 variance fraction, seed 42 and a maximum of 1,000 iterations. Component review used EOG association, scalp maps and time-course inspection. Component 0 was removed from each participant's own fit. For sub-002, component 1 was retained after review. Across the 120 delay windows per participant, the median absolute Fp2 correlation with vertical EOG went from 0.896 to 0.059 for sub-001 and from 0.307 to 0.081 for sub-002.

Table 5: Final peak-to-peak screening of corrected delay windows. The amplitude columns summarize the maximum across EEG channels in each trial.

Participant	Trials	Median (μV)	Highest (μV)	Trials $> 500\,\mu$V
sub-001	120	67.458	142.577	0
sub-002	120	66.930	249.077	0

No channel within a trial fell below the flatness threshold of 0.1 μV peak-to-peak. All 240 trials were retained. The results are recorded in `final_pilot_signal_quality.csv`.

3.6 Tabular Cleaning of the Feature Table

The corrected feature audit found zero missing, infinite or negative band-power values, zero duplicate participant/trial keys and zero duplicate full rows. No imputation was needed. For each participant and feature, values outside the interval from $Q_1 - 1.5\,\mathrm{IQR}$ to $Q_3 + 1.5\,\mathrm{IQR}$ were flagged for review. At least one of 57 features was flagged in 56 trials for sub-001 and 57 for sub-002. All flagged rows were retained. The 15 response-review flags were kept separately from the feature flags.

4 Phase 2: Feature Strategy

4.1 Spectral Features

We used the corrected delay-to-probe windows for all 19 EEG channels. Welch power spectral density [1] was computed with a Hann window, 2 s segments, 1 s overlap, constant detrending and density scaling. Signals were expressed in microvolts before estimating the spectrum. Trapezoidal integration over theta (4–8 Hz), alpha (8–13 Hz) and beta (13–30 Hz) included both

frequency endpoints. Each trial produced $19 \times 3 = 57$ band-power features in μV^2. The table `eeg_features_cleaned_signal.csv` also records trial timing, condition and signal-quality metadata.

4.2 Scaling and Encoding

For the scaling demonstration, the 57 band-power columns were standardized with means and population standard deviations estimated from sub-001 and applied to both participants. There were no constant reference features. The median feature mean and standard deviation were 0 and 1 for sub-001 and -0.2907 and 0.8794 for sub-002. The resulting table is `eeg_features_scaled_demo.csv`. Future evaluations will fit scaling parameters within each training split.

Conditions 5R, 6R, 7R, 5M, 6M and 7M map to nominal IDs 0 through 5, respectively. Set size (5, 6 or 7) and task type (R or M, encoded as 0 or 1) are also stored in `eeg_features_encoded.csv`. The input feature list contains only band powers. Trial identifiers, labels, response audit fields and quality flags are stored as separate columns.

5 Exploratory Data Analysis

5.1 Exploratory analysis of the two-participant pilot

Each participant contributed 20 trials to each of the six conditions, giving 40 trials per condition and 240 in total.

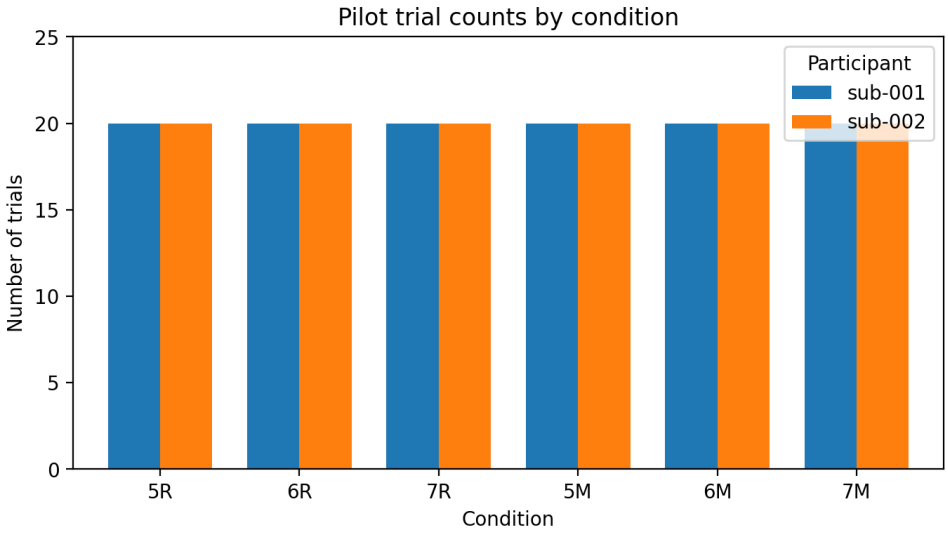


Figure 1: Trial counts by condition and participant.

For each trial and frequency band, we summarized delay-period power using the median across 19 EEG channels. Sub-001 had higher median theta and beta power. Sub-002 had slightly higher median alpha power and a wider alpha interquartile range.

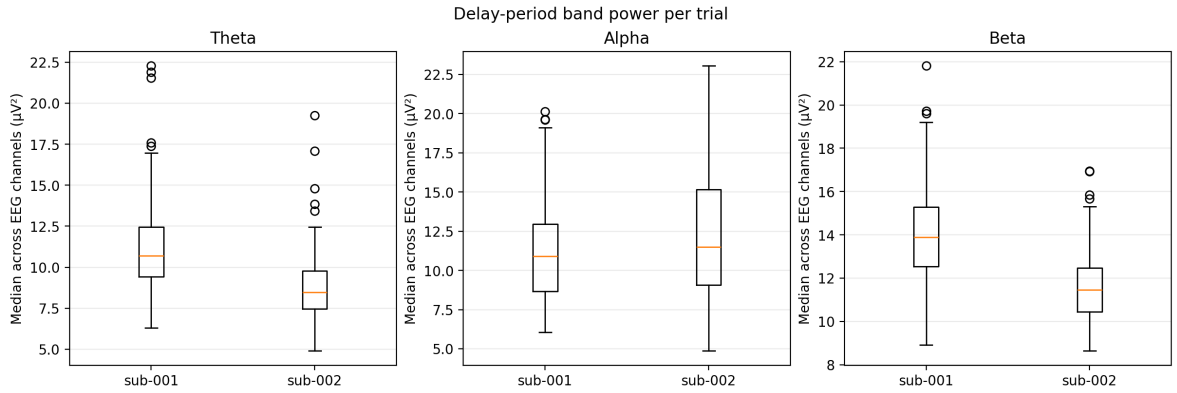


Figure 2: Distribution of trial-level median EEG channel power by band and participant.

The Pearson heatmap showed groups of positive correlations within bands. Beta also contained groups of negative correlations between channels. Correlations between bands varied in sign. The analysis pools trials from both participants, so recording differences can contribute to these patterns.

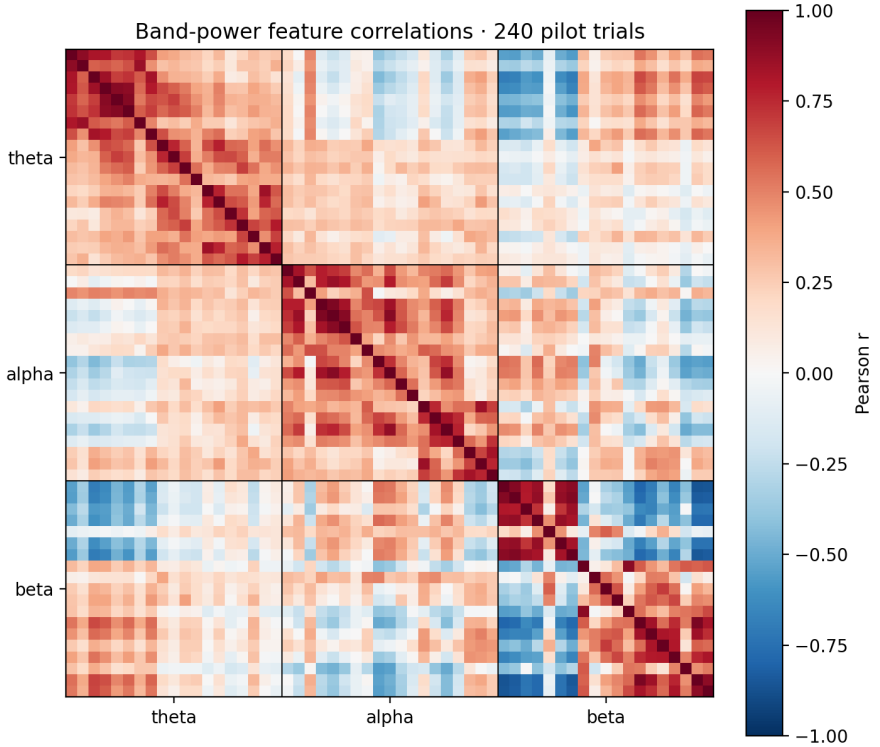


Figure 3: Pearson correlations among 57 unscaled band-power features across 240 pilot trials.

5.2 Response-marker review

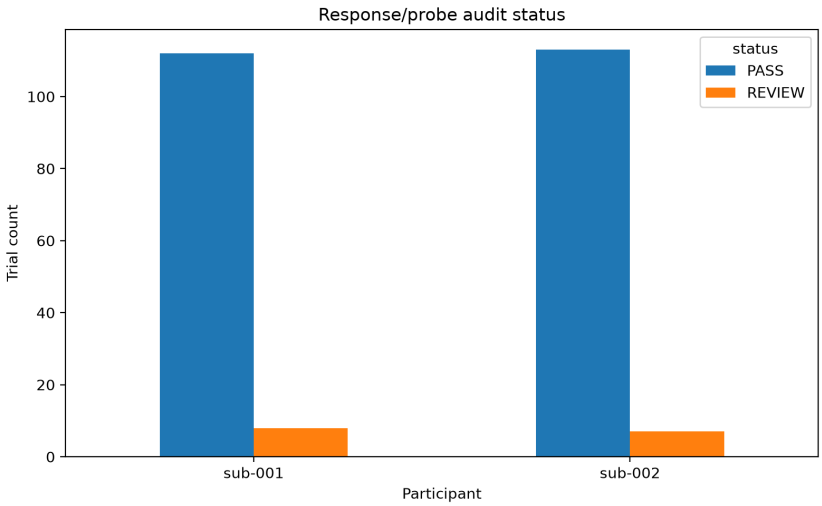


Figure 4: Response-marker review status by participant.

Of 240 probes, 225 passed the structural audit and 15 required review (eight for sub-001 and seven for sub-002).

6 Limitations of the Initial Pilot

1. Findings describe two pilot recordings. The expanded cohort is described in Phase 3.
2. Response accuracy remains provisional for ambiguous markers. The condition labels used here come from the experimental task markers.
3. The cause of the pre-task sub-001 boundary values is unconfirmed. Amplitude and flatness thresholds are screening heuristics.
4. The pilot ICA fits used each participant's full task crop. Future evaluations must define the ICA fitting or calibration procedure relative to the evaluation split, alongside training-only scaling.
5. Response-label validation remains open. Phase 3 and Phase 4 describe the expanded cohort and the model comparison.

7 Reproducibility

The repository contains the acquisition and audit scripts under `data_scripts/`. The Colab notebook `notebooks/EEG_Preprocessing_Pipeline.ipynb` records the pilot signal correction, feature extraction, encoding, audit, scaling demonstration and EDA. Reviewed ICA solutions, corrected recordings and derived tables are saved to Drive checkpoints. Raw EEG stays outside the repository. Repository: `https://github.com/EmanAdil122/federated-eeg-working-memory`.

8 Phase 3: Model Architecture and Implementation

8.1 Expanded cohort and prediction target

The two-participant pilot workflow was extended to 24 participants from OpenNeuro `ds003655`. The merged table contained 2,880 trials. Two trials with abrupt residual EEG amplitudes above the 500 μV review threshold were excluded from the primary analysis (sub-014 trial 62 and sub-020 trial 53), leaving 2,878 trials. The classifier uses 57 delay-period EEG band-power features and predicts memory load (5, 6 or 7 letters) across retention (R) and manipulation (M). The target comes from task-condition markers. Response correctness and response-review status were excluded from the model inputs.

8.2 Shared classifier and training strategies

We implemented a multilayer perceptron in PyTorch with 57 inputs, hidden layers of 64 and 32 units with ReLU activations, dropout of 0.4 after the first hidden layer, and three output logits. Training used cross-entropy and Adam. The features were standardized using the means and standard deviations estimated from the 16 training participants and applied to validation and test features. All strategies used the same feature scale.

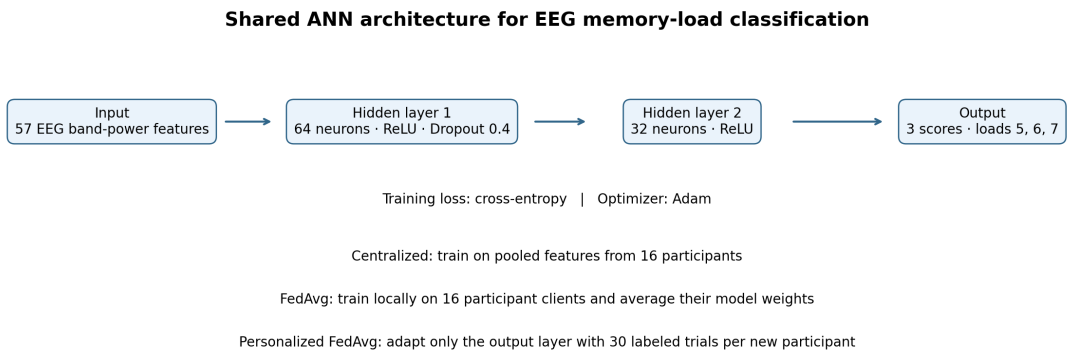


Figure 5: Shared architecture and the three training strategies.

- **Centralized ANN.** Pooled the training participants' features and trained with mini-batches of 64 for 30 epochs. The checkpoint with the best validation macro-F1 was selected.
- **FedAvg ANN.** Simulated 16 participant clients in one runtime. Each client started each round from the current global weights, trained locally for one epoch, and contributed weights to the server average in proportion to its trial count [3]. All clients participated in each of the 30 rounds. The architecture and optimizer settings were carried over from centralized tuning.
- **Personalized FedAvg ANN.** Initialized from the selected global FedAvg checkpoint. For each evaluation participant, the output layer was adapted with Adam using 30 labeled calibration trials, the first five primary trials in each of the six conditions. The hidden layers were frozen. Calibration used full-batch updates, a learning rate of 0.001 and 10 epochs selected on validation participants.

8.3 Hyperparameter selection

Table 6 reports the best validation checkpoint for each centralized configuration. The batch size was held fixed at 64. Dropout of 0.4 was selected because it gave the highest validation

macro-F1 (0.4010). The FedAvg checkpoint at round 8 reached 0.3014 macro-F1 and 32.29% validation accuracy on the full validation set.

Table 6: Centralized ANN tuning. Checkpoints were ranked by validation macro-F1.

Configuration	Hidden units	Dropout	Learning rate	Epoch	Accuracy	Macro-F1
Baseline	64, 32	0.2	0.0010	10	0.3833	0.3829
Smaller network	32, 16	0.2	0.0010	12	0.3833	0.3791
More dropout	64, 32	0.4	0.0010	10	0.4042	0.4010
Lower learning rate	64, 32	0.2	0.0005	11	0.3938	0.3796

Personalization was tuned on held-out validation trials after calibration. Mean participant macro-F1 for 0, 1, 3, 5 and 10 adaptation epochs was 0.2518, 0.2540, 0.2682, 0.2819 and 0.2986, respectively. These scores use a different validation subset and aggregation from the full-set FedAvg checkpoint score.

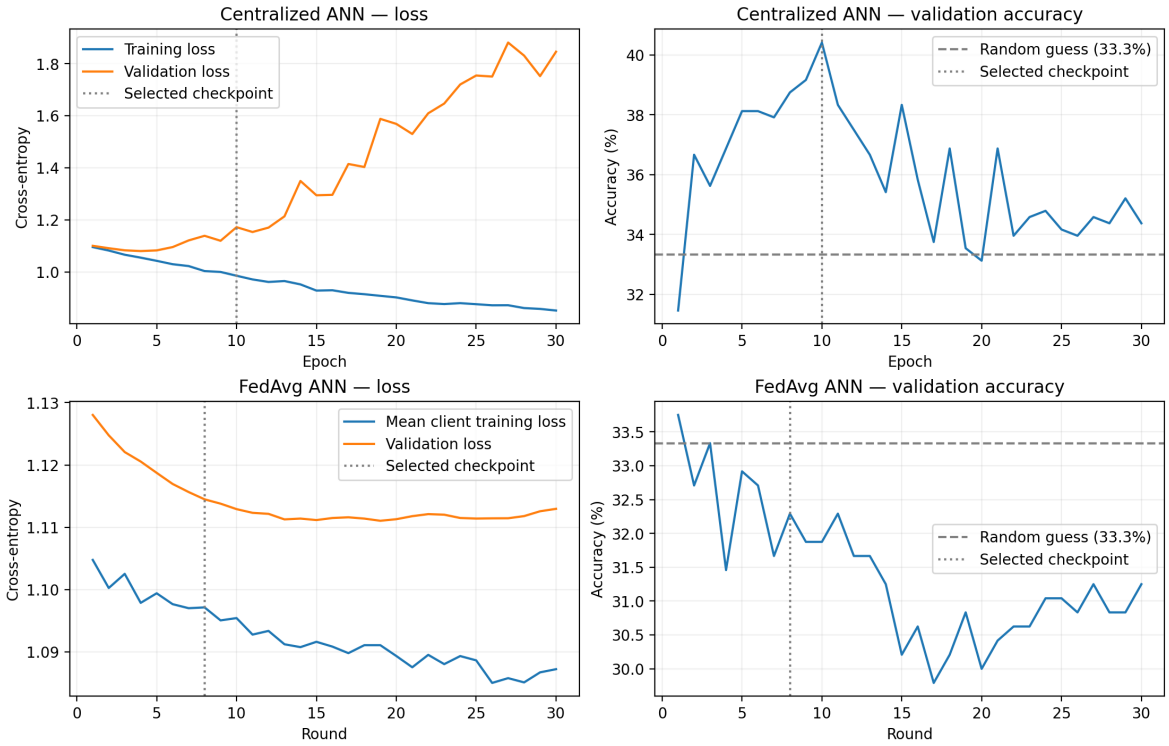


Figure 6: Centralized loss and validation accuracy by epoch, and FedAvg client training loss and validation accuracy by round. Dotted vertical lines mark checkpoints selected by validation macro-F1.

The centralized training loss decreased while validation loss increased in later epochs, consistent with overfitting on this split. FedAvg validation accuracy stayed near or below the uniform-guess reference for most rounds.

9 Phase 4: Training and Empirical Evaluation

9.1 Participant split and test protocol

Participants were assigned to training (16), validation (4) and test (4) using seed 42. No participant was assigned to more than one split. Test participants were sub-002, sub-009, sub-014 and

sub-022. Validation results determined model checkpoints and adaptation settings. Of the test participants' 479 primary trials, 120 were reserved for personalized calibration (30 per participant). The three models were scored on the same remaining 359 trials. Centralized and FedAvg weights were fixed for testing, while personalization used only the labeled calibration subset for the respective test participant. Calibration and evaluation trials came from the same recording session.

Accuracy measures overall correctness. Macro precision, recall and F1 give each load class equal weight. Macro one-versus-rest ROC-AUC summarizes how well predicted probabilities rank each class against the other two classes, independently of the final class decision.

9.2 Held-out results

Table 7: Final test performance on 359 matched trials from four participants.

Model	Accuracy	Macro precision	Macro recall	Macro-F1	Macro ROC-AUC
Centralized ANN	0.3036	0.3098	0.3035	0.3036	0.4752
FedAvg ANN	0.3538	0.3573	0.3543	0.3276	0.4750
Personalized FedAvg ANN	0.3538	0.3643	0.3541	0.3397	0.4797

FedAvg achieved higher accuracy and macro-F1 than the centralized model on this held-out split. Personalization kept accuracy at 35.38% and raised macro-F1 from 0.3276 to 0.3397. Accuracy remained close to the 33.33% uniform-guess reference, and macro ROC-AUC ranged from 0.4750 to 0.4797. These results provide limited evidence for reliable load decoding or consistent superiority on unseen participants.

9.3 Error Analysis

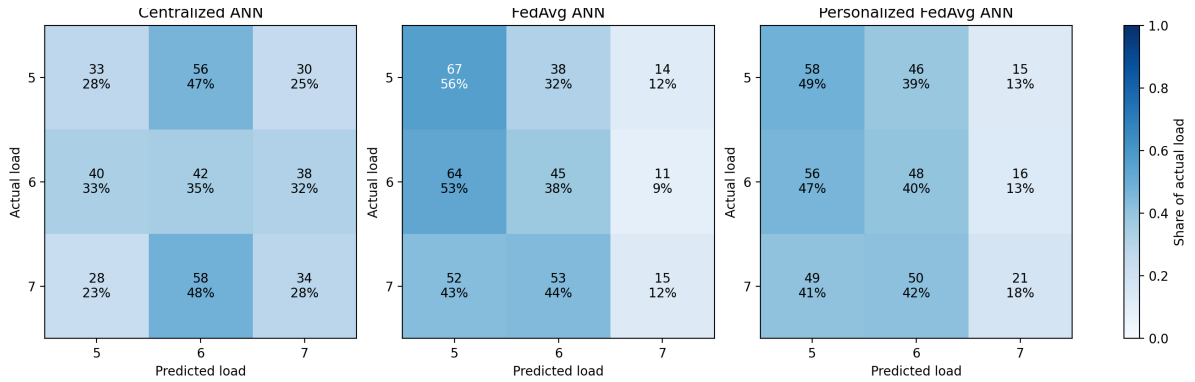


Figure 7: Test confusion matrices. Rows are actual loads and columns are predicted loads. Each cell shows the count and percentage within the actual class.

The centralized model predicted load 6 on 156 of the 359 trials. FedAvg predicted load 5 on 183 trials. Load 7 was particularly difficult for both federated models. FedAvg correctly classified 15 of 120 load-7 trials (12.5% recall); personalization raised this to 21 of 120 (17.5%). Personalization also improved load-6 recall from 37.5% to 40.0%, while load-5 recall dropped from 56.3% to 48.7%.

9.4 Bias Check

Table 8: Descriptive test macro-F1 by participant and task.

Group	Centralized	FedAvg	Personalized FedAvg
sub-002	0.350	0.340	0.371
sub-009	0.242	0.303	0.317
sub-014	0.209	0.243	0.281
sub-022	0.241	0.270	0.290
Manipulation (M)	0.273	0.316	0.292
Retention (R)	0.325	0.330	0.383

Personalization improved participant-level macro-F1 relative to FedAvg for each of the four test participants. The task-level comparison differed: retention improved from 0.330 to 0.383, while manipulation decreased from 0.316 to 0.292. These comparisons describe performance variation within the available split. Demographic fairness was not assessed in this experiment.

9.5 Limitations

The evaluation uses one split and only four test participants. ICA correction was fitted separately on each participant's full task recording before model splitting. This protocol assumes access to the full recording for preprocessing and does not establish prospective performance on an incoming trial stream. Personalization uses labeled trials from each test participant's session, so it measures adaptation to that session. FedAvg was a single-runtime simulation using a shared scaler estimated from training participants. It does not demonstrate a deployed privacy system.

10 Contribution Table

Table 9 summarizes the work recorded on the repository's `main` branch. The EEG validation and exploratory analysis in Deliverable 1 provided the foundation for the modeling and evaluation in Deliverable 2.

Table 9: Team contributions and representative Git commits.

Member	Git author	Work and commit evidence
Eman Adil	`EmanAdil122`	EEG acquisition and initial validation (`6134493`); event and trial-quality audit scripts (`4b57ee1`); pilot feature extraction and signal-quality analysis (`364382c`).
Dua Sarfraz	`duasarfarz3114`	Problem definition (`d3b0627`); band-power EDA script and plot refinements (`64e25ca`, `7dc6a6a`); tabular cleaning outputs (`a5af83f`); pilot report revisions (`e08571d`).
Ayza Ahmad	`ayzaahmed27`	Batch EEG preprocessing and quality review for 24 participants (`53df3a5`); cohort EDA (`3304e5e`); modeling and evaluation notebook (`49fbb20`); runnable training, FedAvg, personalization and evaluation pipeline, with dependencies and run instructions (`2cd5f34`); Deliverable 2 report and evaluation figures (`bf650af`).

References

[1] P. Welch, "The use of fast Fourier transform for the estimation of power spectra," *IEEE Trans. Audio Electroacoust.*, 15(2):70–73, 1967.

[2] A. Gramfort et al., "MEG and EEG data analysis with MNE-Python," *Front. Neurosci.*, 7:267, 2013.

[3] B. McMahan et al., "Communication-efficient learning of deep networks from decentralized data," *AISTATS*, 2017.